

The Emergence of Quantum Mechanics from Deterministic Dynamics: A Mathematically Rigorous Pathway with Critical Missing Proofs

Mathematical Foundations of the Microscopic Substrate

The foundational premise of Invariant Set Theory (IST) posits that the universe is a deterministic dynamical system evolving precisely upon a specific geometric structure within its state space ^{87 405}. This structure, denoted as the Cosmological Invariant Set IU , is conjectured to be a non-computable, fractal-like subset ^{84 174}. This ontological claim has profound and immediate consequences for the mathematical frameworks required to describe the microscopic dynamics, creating a significant departure from standard quantum mechanical formulations which typically operate on a smooth Hilbert space. The primary challenge stems from the intrinsic geometric and topological properties of a fractal set, particularly its singularity with respect to Lebesgue measure and the non-differentiability of trajectories it supports.

A fractal set, by definition, possesses a complex, self-similar structure at arbitrarily small scales and often exhibits a non-integer Hausdorff dimension, distinguishing it from conventional smooth manifolds ³⁸⁸⁵⁷⁰. In the context of IST, the invariant set IU is described as having zero Lebesgue measure, meaning it occupies no volume in its ambient state space ^{84 405}. This property immediately places the entire physical reality outside the domain of classical differential geometry and calculus. The very concept of a derivative, which relies on the ability to probe a function's behavior in an infinitesimal neighborhood around a point, becomes problematic when that point resides on a set with no interior points. The dynamics are confined to this sparse, intricate web of points, suggesting that any attempt to define a velocity field microscopically must confront this fundamental lack of local smoothness. Some analyses of IST suggest a connection between the geometry of IU and the algebraic structure of p-adic numbers, proposing that the fractal set is homeomorphic to the set of p-adic integers ⁸⁹. Since the p-adic integers are topologically equivalent to the Cantor set, this implies that the mathematics

of non-Archimedean fields might provide a more natural language for describing the invariant set's properties than standard real analysis ⁵⁶³⁵⁶⁵.

This singular, non-smooth geometry directly impacts the nature of particle trajectories. In classical mechanics, a trajectory is a smooth curve parameterized by time. On a fractal invariant set, such a description is untenable. Instead, the evolution of a system's state is constrained to move along paths that lie entirely within IU . These paths may exhibit wild oscillations or other irregular behavior, leading to the possibility of non-differentiable motion even if the path is continuous. Research into non-differentiable Bohmian trajectories confirms this concern; it has been shown that while the integral curves of a limiting velocity field may converge to yield meaningful trajectories, the velocity field itself may not exist in a conventional sense at every point ^{11 14}. This presents a critical obstacle: the Bohmian guidance equation, which defines the velocity field, presupposes a level of differentiability that may be absent in the microscopic theory. If the phase function S associated with the wave function is not sufficiently regular, the expression ∇S becomes ill-defined, undermining the very foundation of the guidance equation ³⁹. Therefore, the first major hurdle is to understand how a velocity can be meaningfully defined or approximated on a set where standard differentiation fails.

To navigate this challenging mathematical terrain, one must turn to specialized branches of mathematics. Ergodic theory provides the essential vocabulary for studying the statistical properties of dynamical systems over long timescales ^{50 55}. An ergodic system is one where, in a statistical sense, time averages of observables along a single trajectory equal the space average over the entire state space with respect to an invariant measure μ ¹⁵³¹⁵⁶. For IST, the existence of an invariant measure μ on the fractal set IU is a prerequisite for making any statistical predictions. This measure μ is not the Lebesgue measure, as IU has zero volume; rather, it is a probability measure supported exclusively on the points of the invariant set ¹⁹³¹⁹⁸. The study of such measures on Cantor sets and other fractal geometries is a well-established field, connecting concepts from measure theory, topology, and dynamical systems ^{49 51 167}. The focus shifts from pointwise properties, which are undefined, to properties of integrals against this singular measure.

Another crucial area of mathematics is the analysis of flows and vector fields on metric-measure spaces, which generalize the notion of Riemannian manifolds ²³². In these spaces, one can define notions of gradient, divergence, and flow even when a traditional coordinate system or metric tensor is not available. Dirichlet form theory, for instance, offers a powerful framework for defining Laplacians and associated stochastic processes on fractal-like spaces ¹⁶⁷. The development of a fluid-like picture, where the Bohmian velocity field acts as a flow on a configuration space, suggests that such tools could be

instrumental ¹²². The challenge remains to connect these abstract constructions to the specific geometric properties of the IST invariant set. The process of "stretching and folding" that generates fractal attractors in chaotic systems, such as the Lorenz attractor, provides a physical intuition for how complexity can arise from simple deterministic rules ^{20 334}. However, translating this intuitive understanding into a rigorous derivation of a continuum velocity field requires a precise mathematical formulation of how these singular geometric operations average out over many scales to produce smooth macroscopic behavior. The entire endeavor hinges on the validity of a scale separation assumption, where the Planck-scale structure of IU gives rise to observable physics at much larger scales. This assumption underpins all of coarse-graining theory, from statistical mechanics to renormalization group methods, but its justification in the context of IST is purely physical and lacks a rigorous mathematical proof based on the postulated dynamics ^{9 111}. Without such a proof, the transition from the singular microscopic world to the smooth macroscopic one remains an unproven step.

The Macroscopic Target: Regularity and Uniqueness of the Bohmian Velocity Field

The objective of the coarse-graining procedure is to derive a macroscopic description characterized by a density ρ , a current J , and a velocity field $v=J/\rho$ that satisfies the Bohmian guidance equation, $v=(\hbar/m)\text{Im}(\nabla\psi/\psi)$ ¹²⁶²⁷⁵. Achieving this requires moving from a singular, geometrically rough microscopic substrate to a smooth, analytically well-behaved macroscopic field. The key to this transition lies in establishing sufficient regularity for the emergent velocity field. Regularity, in this context, refers to the degree of smoothness or differentiability of the functions involved, and it is a critical requirement for ensuring the uniqueness and stability of solutions to the governing equations.

For the Bohmian guidance equation to be meaningful, the wave function ψ must be at least weakly differentiable. The standard Schrödinger equation, which governs the evolution of ψ , often imposes specific regularity conditions on its solutions ⁶¹. For instance, if the potential energy term is sufficiently regular, the solution will have derivatives up to a certain order in the L^p sense, placing it in a corresponding Sobolev space $W^{k,p}$ ^{18 249}. Sobolev spaces are a cornerstone of modern analysis because they allow for the treatment of functions that may not be classically differentiable but whose derivatives exist as generalized or distributional objects ⁴⁰¹. The requirement for the

potential and its derivatives to be bounded often translates into the need for the wave function to belong to a Sobolev space with sufficient regularity to guarantee global existence and uniqueness of the Bohmian particle trajectories ^{17 19}. Thus, a successful coarse-graining process must not only produce a density ρ but also a wave function ψ (from which $\rho=|\psi|^2$) that belongs to an appropriate Sobolev space.

The second, and perhaps more subtle, requirement is the uniqueness of the velocity field. Even if a velocity field can be constructed, it is crucial that it be unique to avoid ambiguity in the macroscopic predictions. In the theory of ordinary differential equations (ODEs), the Picard-Lindelöf theorem guarantees the existence and uniqueness of solutions provided the vector field (in this case, the velocity field) is Lipschitz continuous ⁵⁵⁰. However, many physically relevant systems, including those emerging from singular substrates, may have velocity fields that are not Lipschitz. Osgood's theorem generalizes this result, showing that uniqueness holds if the vector field satisfies a weaker condition known as the Osgood criterion ⁵⁰³⁵⁷⁵. This criterion involves the modulus of continuity of the vector field and allows for some degree of singularity, such as log-Lipschitz behavior ⁵⁹¹. Recent mathematical work has demonstrated the sharpness of this condition: for any modulus of continuity that fails the Osgood criterion, one can construct a divergence-free velocity field for which the associated ODE admits multiple distinct solutions ²⁸¹²⁸²⁴⁹⁹⁵⁰⁰. This highlights a significant potential obstruction. If the coarse-grained velocity field derived from the fractal set fails this criterion, the resulting Bohmian mechanics would not be deterministic in its macroscopic description, contradicting the foundational principles of the theory.

The problem of uniqueness extends to the continuity equation, which describes the conservation of probability (or mass) ^{72 75}. The continuity equation is given by $\partial_t \rho + \nabla \cdot J = 0$, where the current J is related to the velocity by $J = \rho v$ ¹²⁶. When the velocity field is low-regularity, this equation can exhibit non-uniqueness of solutions, even among signed measures ¹⁴¹²⁸⁴. The DiPerna-Lions theory established a framework for proving the existence and uniqueness of a "flow" for vector fields in Sobolev spaces, which in turn ensures the uniqueness of solutions to the associated continuity/transport equation ¹⁴⁵. More recent results have refined these conditions, showing that for an Osgood velocity field, all weak solutions are Lagrangian and therefore unique ³⁴⁰. This connects the problem of trajectory uniqueness back to the uniqueness of the density evolution.

The table below summarizes the hierarchy of regularity conditions and their implications for the uniqueness of solutions to the continuity/transport equation, drawing from the provided literature.

Regularity Condition on Velocity Field	Implication for Uniqueness of Solutions	Key Mathematical Result/Framework
Lipschitz Continuity	Unique solutions exist.	Picard-Lindelöf Theorem 550
Log-Lipschitz Continuity	Unique solutions exist.	DiPerna-Lions Theory 145, Osgood Criterion 592
Osgood Condition (Weaker than Log-Lipschitz)	Unique solutions exist.	Sharpness of Osgood Criterion 281282
Failure of Osgood Condition	Non-uniqueness of solutions can occur.	Construction of Non-Unique Flows 289549
Divergence-Free Vector Field in Sobolev Space ($W^{1,p}$)	Can exhibit non-uniqueness and loss of regularity in the solution.	Examples of Non-Uniqueness 139602

This analysis reveals that the success of the coarse-graining program is contingent on the limiting velocity field possessing a level of regularity that is stronger than mere continuity. The failure to establish such regularity would mean that the same microscopic theory could give rise to multiple, inequivalent macroscopic quantum theories, rendering the derivation non-unique and therefore incomplete. The challenge, therefore, is twofold: first, to prove that the coarse-graining process produces a velocity field with sufficient regularity, and second, to confirm that this regularity is enough to satisfy a criterion like the Osgood condition, thereby ensuring the uniqueness of the resulting Bohmian dynamics.

The Coarse-Graining Bridge: From Singular Measures to Smooth Hydrodynamics

The central task of connecting the singular microscopic dynamics on the invariant set IU to the smooth macroscopic Bohmian velocity field necessitates a mathematically rigorous coarse-graining mechanism. This bridge cannot rely on ad hoc assumptions but must be grounded in a formal mathematical framework capable of handling the transition from a discrete or fractal distribution of states to a continuous fluid-like description involving densities and currents. The literature on hydrodynamic limits, empirical measure convergence, and optimal transport theory provides several promising candidates for this bridge, each offering a different perspective on how a large number of microscopic particles can collectively behave like a continuous medium.

A foundational concept in this context is the weak convergence of measures 12 91 . In statistical mechanics, one often starts with a system of N particles and considers the

empirical measure, which is a sum of delta functions centered on the instantaneous positions of the particles. As N tends to infinity, one hopes that this discrete measure converges weakly to a smooth density function $\rho(x)dx$. Weak convergence means that the integral of any continuous, compactly supported test function against the empirical measure approaches its integral against the limit measure. This approach is used to justify the passage from a microscopic particle description to a macroscopic continuum description ²⁷⁸⁵³⁹. However, as explicitly noted in the research protocol, weak convergence of the density alone is insufficient. It establishes the existence of a limit density ρ but provides no information about the existence, regularity, or uniqueness of a velocity field or current ¹². The dynamics, which are encoded in the evolution of the measure, are lost in the process of passing to the limit.

To capture the dynamics, one must consider not just the density but also the flux or current. A more powerful approach involves analyzing the convergence of the empirical current ²¹⁷⁴⁴⁶. If the microscopic system has a well-defined velocity at each point, one can construct a current density $J_\epsilon = \rho_\epsilon v_\epsilon$ at a coarse-graining scale ϵ . The goal is then to show that as $\epsilon \rightarrow 0$, the sequence of current densities J_ϵ converges in a suitable topology to a limit current J . Convergence could be established in a dual space, such as the space of distributions or a negative-order Sobolev space, where it is tested against smooth vector fields ²⁴⁷. If both the density ρ_ϵ and the current J_ϵ converge, then their ratio, the velocity field $v_\epsilon = J_\epsilon / \rho_\epsilon$, can be analyzed, provided the density does not vanish. This two-step process—convergence of both density and current—is a more robust method for deriving a hydrodynamic limit.

A particularly potent framework for formalizing this coarse-graining process is optimal transport theory ⁴³⁹⁵³⁷⁶²³. Optimal transport provides a way to define a geometry on the space of probability measures, using the Wasserstein distance, which quantifies the "cost" of transforming one measure into another ³⁶⁴. In this view, the evolution of the particle density ρ_t from an initial time to a final time can be understood as a geodesic in this infinite-dimensional space of measures. The velocity field v_t that transports the initial measure to the final measure is the tangent vector to this geodesic. This perspective naturally leads to the continuity equation, as the requirement for the measure to be transported without creation or destruction of mass is precisely what the continuity equation enforces ⁶²³. This framework is highly relevant because it connects the evolution of the macroscopic density directly to the properties of the underlying velocity field, providing a principled way to define the coarse-graining map $\mathcal{C}_\epsilon$. It allows for the treatment of singular initial data and can handle cases where the velocity field is not smooth, making it a promising tool for bridging the gap between the singular IST substrate and the smooth Bohmian target ⁵³⁴.

Furthermore, the theory of measure-valued solutions to partial differential equations, particularly the continuity equation, offers another avenue for analysis ¹⁴⁶²⁸³. In this setting, the density ρ and current J are not necessarily functions but are allowed to be Radon measures. This is useful because even if the limit of a sequence of smooth densities is singular, it can still be represented as a measure. The challenge then becomes to prove that this limiting measure is absolutely continuous with respect to Lebesgue measure, thereby yielding a true density function ρ , and that the corresponding current is also absolutely continuous, yielding a velocity field $v=J/\rho$. The Radon-Nikodym theorem is the key tool here, stating that if one measure is absolutely continuous with respect to another, then a density (the Radon-Nikodym derivative) exists ⁴⁷⁰⁴⁷¹. For example, a Cantor measure is singular with respect to Lebesgue measure, while a Dirac measure is pure point ⁴⁶⁸. The coarse-graining process must effectively "smooth out" the microscopic measure μ into one that is absolutely continuous, producing a physically sensible macroscopic description. The convergence of the microscopic currents to a macroscopic current can be studied using techniques from compensated compactness, which deals with the weak convergence of nonlinear expressions like $F(v_j)$ ⁴⁴⁴⁴⁴⁵. By combining these advanced tools from PDE theory and measure theory, a mathematically sound pathway from the discrete, singular microscopic world to the continuous, differentiable macroscopic world can be formulated.

Existence and Properties of the Invariant Set Measure

Before any coarse-graining can proceed, a fundamental prerequisite must be met: the existence and characterization of a physically meaningful invariant measure μ on the fractal invariant set $I\mathcal{U}$. In a deterministic dynamical system, an invariant measure is a measure that is preserved by the system's evolution operator ¹⁹²¹⁹⁹. For IST, this measure μ is supposed to describe the statistical distribution of states that the universe passes through over time. Without such a measure, the system is merely a collection of trajectories with no statistical content, making it impossible to perform the thermodynamic or hydrodynamic limits required to connect to macroscopic physics. The challenge in the context of IST is that $I\mathcal{U}$ is a singular set of zero Lebesgue measure, so the standard Liouville measure cannot be used. The relevant measure μ must be singular with respect to Lebesgue measure.

In the broader field of dynamical systems, a specific type of invariant measure, known as a Sinai-Ruelle-Bowen (SRB) measure, plays a crucial role in describing the statistical behavior of chaotic systems ¹³¹²²⁴. SRB measures are named after Yakov Sinai, David

Ruelle, and Rufus Bowen, who developed the theory for a class of hyperbolic dynamical systems ¹³⁵. A key property of an SRB measure is that it describes the statistics observed along typical orbits (with respect to Lebesgue measure) in the system's basin of attraction. While the SRB measure itself is often singular with respect to Lebesgue measure, its conditional measures on unstable manifolds are absolutely continuous ¹²⁹⁵⁶⁰. This mixed character makes SRB measures exceptionally well-suited for describing systems with strange attractors, which are precisely the kind of fractal structures seen in chaotic dynamics ¹³⁴³⁸³. Given that the IST invariant set IU is explicitly a fractal, conjecturing that its physical measure μ is an SRB measure is a natural and powerful hypothesis. This conjecture would place IST on firm ground within the well-developed mathematical theory of chaotic systems.

The existence and uniqueness of SRB measures have been proven for a wide range of dynamical systems, particularly for uniformly hyperbolic (Axiom A) systems and certain classes of non-uniformly hyperbolic systems ¹²⁸²²⁵³⁰⁰. For example, for the geometric Lorenz attractor, a canonical example of a strange attractor, the existence and uniqueness of an SRB measure is well-established ³⁰⁰. Similarly, for singular hyperbolic attractors in higher dimensions, existence and uniqueness have been proven ¹²⁸. The mathematical machinery developed for these systems relies on constructing a Markov partition and analyzing the induced dynamics on a symbolic space, ultimately proving the existence of an equilibrium state that corresponds to the SRB measure ²⁹⁹³⁶¹. If the dynamics of IST were known to possess similar hyperbolic properties, one could invoke these existing theorems to guarantee the existence of a unique SRB measure μ .

However, a critical issue arises in the specific context of IST. The invariant set IU is not just any strange attractor; it is conjectured to be a non-computable fractal ^{84 174}. The known existence theorems for SRB measures typically require certain regularity and hyperbolicity conditions on the dynamical system that may not be applicable to a system whose state space is a non-computable set. The assumptions needed to prove the existence of an SRB measure—for example, uniform expansion and contraction rates—are not established for the IST model; they remain hypotheses ¹²⁸. Therefore, while the SRB measure provides a compelling candidate for the physical measure μ , its existence and uniqueness for the specific IST dynamics is currently an open problem. Establishing this would be a monumental step forward, as it would provide the rigorous statistical foundation necessary for any subsequent coarse-graining argument. It would allow one to leverage the vast body of literature on the statistical properties of SRB measures, such as their mixing properties, decay of correlations, and the applicability of limit theorems like the Central Limit Theorem ^{93 94 157}. These properties are essential for justifying the emergence of irreversible thermodynamic behavior and stable macroscopic laws from an

underlying reversible microscopic dynamics, a key feature of any successful emergence story.

In summary, the search for the microscopic measure μ leads directly to the concept of SRB measures. The existence of such a measure is a non-trivial mathematical question that separates plausible models from those that are rigorously founded. For IST, the status of this question is currently unknown. The program stands at the frontier of mathematical physics, requiring new proofs that adapt the powerful theory of SRB measures to the novel and challenging setting of a non-computable fractal state space.

Critical Obstructions and Missing Proofs

Despite the existence of promising mathematical pathways, the ambitious goal of deriving a unique Bohmian velocity field from a fractal invariant set is fraught with significant mathematical obstacles and rests on several critical, yet-to-be-proven assumptions. A thorough investigation reveals that the journey from a singular geometric substrate to a smooth analytical target is not guaranteed and faces serious challenges related to the existence of a physical measure, the regularity of the limiting velocity field, and the potential for non-uniqueness in the resulting dynamics.

The first and most fundamental missing piece of the puzzle is a rigorous proof of the existence and uniqueness of an SRB measure for the IST dynamical system. As detailed previously, an SRB measure serves as the crucial link between the microscopic dynamics on the fractal set IU and its macroscopic statistical properties [131560](#). While SRB measures are well-understood for many classes of chaotic systems, these results are predicated on specific mathematical conditions, such as uniform hyperbolicity or partial hyperbolicity, which are not established for the abstract, non-computable dynamics of IST [128300](#). The conjecture that the physical measure μ is an SRB measure is a vital hypothesis, but without a theorem to support it, the entire statistical foundation of the coarse-graining argument remains speculative. The absence of such a proof represents a major gap, as it leaves open the question of whether the system even possesses a well-defined statistical description that could serve as the basis for a hydrodynamic limit.

Assuming for a moment that an SRB measure μ exists and is unique, a second, equally formidable challenge emerges: the regularity of the limiting macroscopic velocity field. The process of coarse-graining is intended to act as a "smoothing" operation, averaging out fine-scale fluctuations on the fractal set to produce a smooth, differentiable field at

larger scales. However, mathematical analysis of transport and continuity equations with low-regularity velocity fields shows that this smoothing effect is not guaranteed. In fact, it has been demonstrated that even for divergence-free velocity fields that are uniformly bounded in all Sobolev spaces $W^{1,p}$, the solution to the continuity equation can lose all Sobolev regularity and fail to belong to any Sobolev space itself [139291519](#). Another result shows that a passive scalar being transported by a non-Lipschitz velocity field can experience "norm inflation," leading to a complete loss of regularity in Sobolev norms [619](#). This is a grave concern for the IST program. It suggests that the inherent singularity of the microscopic dynamics might be too severe, preventing the coarse-graining process from producing a sufficiently regular velocity field to define a unique Bohmian mechanics. The derivation would fail if the limiting field lacks the minimal regularity required for uniqueness, such as satisfying an Osgood-type condition [281282](#).

This leads to the third major potential obstruction: non-uniqueness. If the limiting velocity field is not regular enough, the continuity equation for the density ρ may admit multiple distinct solutions [297](#). This phenomenon, sometimes called "wild solutions," has been rigorously constructed using methods like convex integration [289599](#). In such a scenario, the coarse-graining process would not be well-defined; a single microscopic theory could, in principle, lead to different macroscopic behaviors. This would render the emergence of a unique quantum theory impossible, as the predictions would depend on an arbitrary choice of solution branch. This is not merely a theoretical curiosity; it is a direct consequence of relaxing the regularity assumptions on the vector field. Any successful derivation must therefore include a proof that the limiting velocity field is "good enough" to preclude such pathological behavior.

The table below outlines these critical gaps and obstructions.

Category	Missing Proof / Assumption	Potential Obstruction / Consequence
Existence of Physical Measure	Theorem proving existence and uniqueness of an SRB measure for IST dynamics.	Without a physical measure, the system lacks a statistical description, making a hydrodynamic limit impossible.
Regularity of Limiting Field	Proof that the coarse-grained velocity field possesses sufficient regularity (e.g., satisfies an Osgood condition).	The velocity field may be non-unique, or the continuity equation may admit multiple solutions, invalidating the derivation.
Uniqueness of Dynamics	Proof that the limiting dynamics are uniquely determined by the microscopic theory.	The emergence of multiple, inequivalent macroscopic theories from a single microscopic theory.
Scale Separation	Rigorous justification of the separation of scales from the Planck scale to the macroscopic scale.	The coarse-graining map may not be well-defined, or the limit may be trivial.

In conclusion, the path from the microscopic invariant set to the macroscopic Bohmian field is blocked by these formidable mathematical hurdles. The program's success is not

merely a matter of finding the right formula but depends on proving several deep and non-trivial results in the theory of dynamical systems and PDEs. The absence of these proofs means that the derivation, while plausible, remains an open question.

Synthesis and Final Verdict

After a comprehensive examination of the mathematical foundations, macroscopic targets, coarse-graining mechanisms, and critical obstructions, a clear verdict can be rendered on the central research question: Can the microscopic dynamics of a deterministic invariant set $(IU, \mathcal{T}, \mu)$ be rigorously coarse-grained into a unique macroscopic Bohmian velocity field?

The analysis reveals that the proposed emergence is neither definitively possible nor demonstrably impossible with the current state of mathematical knowledge. The core difficulty lies in bridging the chasm between the singular, non-differentiable geometry of the fractal invariant set and the smooth, differentiable structure required for the Bohmian guidance equation. While several powerful mathematical frameworks offer potential pathways, each is contingent on assumptions that have not been proven for the specific dynamics of Invariant Set Theory.

On the one hand, there is a coherent and promising strategy for achieving the desired emergence. This strategy rests on three pillars: 1. **Ergodic Theory and SRB Measures:** The use of ergodic theory provides the language of invariant measures for dynamical systems ⁵⁰. The central hypothesis is that the physical measure μ on the IST invariant set is a Sinai-Ruelle-Bowen (SRB) measure ¹³¹. SRB measures are uniquely suited to describe the statistical properties of chaotic systems on fractal attractors, providing a rigorous statistical foundation for the coarse-graining process ⁵⁶⁰. 2. **Continuity Equations and Optimal Transport:** Once an SRB measure is established, the evolution of the system can be described by a continuity equation for the macroscopic density ρ ⁵³⁹. Optimal transport theory offers a sophisticated framework for understanding this evolution as a flow that transports the initial measure to its final state, directly linking the density's evolution to an underlying velocity field ⁴³⁹⁶²³. 3. **Uniqueness via Regularity:** The uniqueness of the resulting Bohmian mechanics would follow if the limiting velocity field possesses sufficient regularity. Criteria weaker than Lipschitz continuity, such as the Osgood condition, are known to be sufficient for the uniqueness of solutions to the continuity equation ²⁸¹²⁸². Therefore, a key goal of the coarse-graining analysis would be to prove that the emergent velocity field satisfies such a condition.

On the other hand, this promising pathway is critically dependent on two major, unproven mathematical statements. The entire derivation collapses if either of these can be shown to be false.

The first and most significant missing proof is the **existence and uniqueness of an SRB measure for the IST dynamical system**. While SRB measures are well-understood for many classes of hyperbolic systems, the abstract and non-computable nature of the IST invariant set means that the standard existence theorems do not apply. This is the foundational gap that must be filled before any further analysis can proceed.

The second critical uncertainty is the **regularity of the limiting velocity field**. Even if an SRB measure exists, there is no guarantee that the coarse-graining process will produce a sufficiently regular velocity field. Mathematical results demonstrate that transport equations can exhibit a catastrophic loss of regularity, even for highly constrained velocity fields, potentially leading to non-unique solutions ¹³⁹⁵¹⁹⁶¹⁹. Without a proof that the limiting field is "well-behaved," the resulting macroscopic theory may not be unique, thus failing the primary objective of the derivation.

Given this balance of strong evidence for a potential pathway and the presence of critical, unresolved mathematical obstacles, the most accurate and responsible verdict is:

Verdict: POSSIBLY, WITH SPECIFIC MISSING PROOF

The BM-IST program has a credible and well-motivated mathematical path toward deriving the Bohmian velocity field, but its success is contingent on proving two specific, non-trivial theorems: one concerning the existence of an SRB measure for the IST dynamics, and another concerning the sufficient regularity of the resulting macroscopic velocity field. Until these proofs are found, the emergence remains a compelling but unproven hypothesis.

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